Fake News Detection Project – Installation & User Guide

1. Introduction This document provides a complete installation guide and user manual for a Fake News Detection system. It explains system requirements, setup instructions, model training steps, and how to use the application after installation.

2. System Requirements **Hardware:**
• 4 GB RAM minimum (8 GB recommended)
• Dual-core processor or above
• 2 GB free storage

Software:
• Windows / Linux / macOS
• Python 3.8 or later
• Pip package manager
• VS Code / PyCharm / Jupyter Notebook

3. Installation Steps **Step 1: Install Python**
Download from the official website and ensure 'Add Python to PATH' is enabled. Verify using:
python --version

Step 2: Create Virtual Environment
python -m venv fakenews-env
Activate the environment based on your OS.

Step 3: Install Required Libraries
pip install numpy pandas scikit-learn nltk flask joblib

Step 4: Download NLTK Data
Inside Python:
import nltk
nltk.download('punkt')
nltk.download('stopwords')

Step 5: Recommended Project Structure
FakeNewsDetection/
■■■ dataset/
■■■ model/
■■■ app.py
■■■ train.py
■■■ preprocessing.py
■■■ templates/
■■■ static/

Step 6: Train the Model
Run:
python train.py
The script should load the dataset, preprocess text, generate TF-IDF features, train a classifier, and save the model as:
model/fake_news_model.pkl

Step 7: Run the Application
Start the Flask server using:
python app.py
Open the application in a browser:
http://127.0.0.1:5000/

4. User Guide **Input:**
Users can enter a news headline, article text, or pasted content.

Processing:
The system cleans the text, removes stopwords, applies TF-IDF vectorization, and predicts using a trained model.

Output:
The system displays:
• Prediction: Fake or Real
• Confidence Score (optional)

5. Example Input:
"Scientists confirm chocolate cures COVID-19."
Output:
Prediction: Fake News
Confidence: 97%

6. Conclusion This guide includes the installation steps and usage instructions needed to set up and operate the Fake News Detection system. The project combines text preprocessing, machine learning models, and a simple web interface to provide reliable predictions.